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SYSTEMS AND METHODS FOR LOW VOLTAGE WIRING FOR EVTOL AIRCRAFT

Abstract

In an embodiment, a system may comprise at least two power supplies which may comprise a first power source, a first step-down converter, a first switching device, a first bus, a second power source, a second step-down converter, a second switching device, and a second bus, the first and second buses being electrically separate. An exemplary system may further comprise an alternate power supply comprising a third power source and an alternate step-down converter connected to at least the first and second buses via at least a third switching device, the alternate power supply acting as a backup power supply configured to be used after a failure of one or more of the at least two power supplies. In some embodiments, the first and second switching devices may be controlled by at least a first controller and the at least third switching device may be controlled by an alternate controller.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to Provisional U.S. Patent Application No. 63/380,376, filed Oct. 20, 2022, which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] This disclosure relates generally to the field of powered aerial vehicles. More particularly, and without limitation, the present disclosure relates to innovations in tilt-motor aircraft that use electrical propulsion systems. Certain aspects of the present disclosure generally relate to power distribution systems for powered aerial vehicles. Other aspects of the present disclosure generally relate to improvements in power distribution systems that may be used in other types of vehicles but provide particular advantages in aerial vehicles.

BACKGROUND

[0003] Extant power distribution systems comprise low-voltage DC buses which are interconnected such that, when a power source failure occurs, power to the buses is maintained and the components utilizing the buses remain powered. Such extant power distribution solutions, however, allow for propagation of noise, overloads, and other electrical disturbances across two or more buses and/or power sources.

[0004] Furthermore, extant power distribution systems comprising low-voltage DC buses each have a primary power source and a backup power source, such as, e.g., a low-voltage battery. Therefore, extant power distribution solutions require extra weight and increased maintenance at least due to the primary and backup power source required for each low-voltage DC bus.

SUMMARY

[0005] Some disclosed embodiments involve a power distribution system for an aircraft. In some embodiments, a power distribution system for an aircraft may comprise at least two power supplies. In some embodiments, at least two power supplies may comprise a first power source connected to a first step-down converter, the first step-down converter being connected via a first switching device to a first bus, and a second power source connected to a second step-down converter, the second step-down converter being connected via a second switching device to a second bus, the second bus being electrically separate from the first bus. In some embodiments, a power distribution system for an aircraft may further comprise an alternate power supply comprising a third power source connected to an alternate step-down converter, the alternate step-down converter being connected to each of at least the first and second buses via at least a third switching device, the alternate power supply acting as a backup power supply configured to be used after a failure of one or more of the at least two power supplies. In some embodiments, the first and second switching devices are controlled by at least a first controller. In some embodiments, the at least third switching device is controlled by an alternate controller.

[0006] Consistent with some disclosed embodiments, a system may be configured to provide alternate power from the alternate power supply within 200 milliseconds after a component of the

at least two power supplies fails. In some embodiments, a system may be configured to provide alternate power from the alternate power supply within 100 milliseconds after a component of the at least two power supplies fails. In some embodiments, a system may be configured to provide alternate power from the alternate power supply within 50 milliseconds after a component of the at least two power supplies fails.

[0007] In some embodiments, the first bus and the second bus may be configured to receive power via the at least third switching device from the alternate power supply after a failure of a component of a channel corresponding to at least one of the first power source or the second power source.

[0008] Consistent with some disclosed embodiments, each one of the first, second, and at least third switching devices may have a conducting state and a non-conducting state, and the at least first controller and the alternate controller may be configured to cause transitions within each of the first, second, and at least third switching devices between their conducting states and their non-conducting states to thereby cause each of the first, second, and alternate power supplies to provide power to each of the first and second buses in different arrangements.

[0009] In some embodiments, the first, second, and at least third switching devices may be configured such that each of the first, second, and alternate step-down converters are electrically separate or electrically separable. In some embodiments, the first, second, and alternate power sources may be high voltage power sources. In some embodiments, the first, second, and alternate step-down converters may be high voltage to low voltage converters. In some embodiments, the first, second, and alternate step-down converters may be DC-DC converters.

[0010] According to some disclosed embodiments, the first and second buses may be configured to be interconnectable. In some embodiments, the first and second switching devices may be configured to disconnect a failing component of a channel corresponding to at least one of the first power source or the second power source from the first bus and/or the second bus.

[0011] Consistent with some disclosed embodiments, the first switching network may be controlled by the at least first controller and the second switching network may be controlled by at least a second controller. In some embodiments, the at least third switching device may include two or more switching devices forming an alternate switching network.

[0012] According to some disclosed embodiments, the third power source may be one of the first power source or the second power source. In some embodiments, the first, second, and third power sources may be the same power source.

[0013] In some embodiments, the first, second, and alternate step-down converters may operate in an absence of a low voltage battery. In some embodiments, the at least first controller may operate based on hardware logic. In some embodiments, the alternate controller may operate based on conditional logic.

[0014] Some disclosed embodiments involve a method for distributing power in an aircraft. In Some embodiments, a method comprises providing at least two power supplies, the at least two power supplies comprising a first power source connected to a first step-down converter, the first step-down converter being connected via a first switching device to a first bus, and a second power source connected to a second step-down converter, the second step-down converter being connected via a second switching device to a second bus, the second bus being electrically separate from the first bus. In some embodiments, a method further comprises providing an alternate power supply comprising a third power source connected to an alternate step-down converter, the alternate step-down converter being connected to each of at least the first and second buses via at least a third switching device, the alternate power supply acting as a backup power supply configured to be used after a failure of one or more of the at least two power supplies. In some embodiments, a method further comprises controlling the first and second switching devices by at least a first controller. Further, in some embodiments, a method may comprise controlling the at least third switching device by an alternate controller.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0015] FIG. 1 shows an exemplary power distribution system, consistent with disclosed embodiments.

[0016] FIG. 2 shows another exemplary power distribution system, consistent with disclosed embodiments.

[0017] FIG. 3 shows an exemplary mode of an exemplary power distribution system, consistent with disclosed embodiments.

[0018] FIG. 4 shows another exemplary mode of an exemplary power distribution system, consistent with disclosed embodiments.

[0019] FIG. 5 shows yet another exemplary mode of an exemplary power distribution system, consistent with disclosed embodiments.

[0020] FIG. 6 shows yet another exemplary mode of an exemplary power distribution system, consistent with disclosed embodiments.

[0021] FIG. 7 shows yet another exemplary mode of an exemplary power distribution system, consistent with disclosed embodiments.

[0022] FIG. 8 shows yet another exemplary mode of an exemplary power distribution system, consistent with disclosed embodiments.

[0023] FIG. 9 shows an exemplary system for powering a low voltage power system, consistent with disclosed embodiments.

[0024] FIG. 10 shows an exemplary method for operating an exemplary power distribution system, consistent with disclosed embodiments.

[0025] FIG. 11 shows another exemplary method for operating an exemplary power distribution system, consistent with disclosed embodiments.

DETAILED DESCRIPTION

[0026] In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the disclosed example embodiments. However, it will be understood by those skilled in the art that the principles of the example embodiments may be practiced without every specific detail. Well-known methods, procedures, and components have not been described in detail so as not to obscure the principles of the example embodiments. Unless explicitly stated, the example methods and processes described herein are neither constrained to a particular order or sequence nor constrained to a particular system configuration. Additionally, some of the described embodiments or elements thereof can occur or be performed (e.g., executed) simultaneously, at the same point in time, or concurrently. Reference will now be made in detail to the disclosed embodiments, examples of which are illustrated in the accompanying drawings.

[0027] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of this disclosure. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate several exemplary embodiments and together with the description, serve to outline principles of the exemplary embodiments.

[0028] This disclosure addresses at least the problems described above by configuring one or more switching networks such that the buses remain electrically separate after any single initial failure of a power distribution system component, including but not limited to the failure of a primary power conversion device (e.g., a converter). A switching network, as used herein, is defined as at least one switching device. The terms “conversion device” and “converter” are used interchangeably throughout this disclosure.

[0029] This disclosure further addresses such problems by providing an alternate/backup power source that is configured to be connectable to all the low-voltage buses via a switching network.

[0030] eVTOL (i.e., electric vertical takeoff and landing) aircraft particularly benefit from the methods and systems disclosed herein because they require electric power for all flight and engine control. Thus, for eVTOL aircraft, the need for high-availability electric power is particularly important.

[0031] Conventional aircraft may have other power sources, such as, e.g., a combustion engine. Conventional aircraft also often comprise flight control systems which may be independent from electric or electronic equipment, such as, e.g., mechanical controls. Furthermore, conventional aircraft which utilize flight control systems with electric or electronic flight controls often also require an emergency power source, such as, e.g., an air-driven generator.

[0032] eVTOL aircraft which utilize the systems and methods disclosed herein may benefit due to (1) high redundancy electric power with minimal equipment, (2) an absence of a need for a dedicated alternate high-voltage battery, a dedicated alternate power converter, and/or a dedicated low-voltage battery for each bus, and (3) an absence of a need for an air-driven generator or other emergency power source.

[0033] In some embodiments, a power distribution system as described herein may comprise a system (or method) for providing high-availability low-voltage power to an aircraft, the power distribution system providing a resultant safety margin significantly higher than currently achievable with requirements based on comparable weight and quantity of equipment. “Low-voltage,” as used herein, refers to 33 volts or lower. For example, a low voltage power distribution system may be responsible for converting high voltage energy from one or more main battery systems, via one or more step-down power converters, into a 28 VDC network which, in turn, may be used to distribute electrical power to low voltage loads of an aircraft. In some embodiments, the system (or method) comprises (providing) two or more high-voltage power source(s), each connected via separate means (i.e., wherein the high-voltage power sources are not interconnected) to a separate primary power conversion device (i.e., power converter) which causes the high-voltage power to be converted to low-voltage direct current (DC) output. It will be understood that, as such, the primary power converter may be a step-down power converter. “High-voltage,” as used herein, refers to a voltage greater than 110 volts. “Low-voltage direct current (DC),” as used herein, refers to 33 VDC or lower. For example, two or more primary and one alternate DC-DC converter(s) may perform the voltage conversion while the power may be managed and distributed through two or more primary power distribution boxes (PDBs) and two or more secondary power distribution assemblies (SPDAs).

[0034] In exemplary embodiments, a low voltage power distribution system may convert high voltage energy into regulated low voltage power to feed all aircraft low voltage loads. For example, the two or more primary step-down converters of the system may provide primary power to the power distribution boxes which, in turn, distribute the primary power among two or more buses. As another example, the alternate converter may act as an alternate step-down converter and thereby an alternate power source and may supply low voltage power during, e.g., power up of an aircraft and/or in the event of a failure of one or more primary converters. In some embodiments, the alternate converter may be connected to at least one primary channel. In some embodiments, at least one battery may be connected to at least one bus, wherein the at least one battery provides backup power to the at least one bus. As another example, the PDBs and SPDAs may distribute low voltage electrical power to all low voltage system loads. As yet another example, the return lines of the low voltage system may be routed among and along the PDBs, SPDAs, and converters. Further, other aircraft system return lines may be routed back to one or more PDBs.

[0035] In other exemplary embodiments, a low voltage power distribution system may provide low voltage electrical power protections, such as identifying operating conditions outside the allowed power quality and capacity envelopes for which the system is designed and taking the necessary actions to remove such conditions. For example, the system may protect against power supply overvoltage by, e.g., continuously monitoring an output voltage to detect any overvoltage and

thereby protect the system against overvoltage conditions. As another example, the system may protect against power supply undervoltage by, e.g., continuously monitoring an output voltage to detect any undervoltage and thereby protect the system against undervoltage conditions. As another example, the system may protect against power supply overcurrent by, e.g., continuously monitoring an output current to detect and remove any overcurrent circulating through the system, based on predefined settings. As yet another example, the system may protect against power supply overload by, e.g., continuously monitoring a used capacity to detect and remove overload conditions, based on predefined settings. As a further example, the system may protect against power supply overtemperature by, e.g., continuously monitoring an internal temperature to detect and report any overtemperature condition. As yet a further example, the system may protect against power distribution overcurrent by, e.g., having each PDB and SPDA supply power to low voltage loads through circuit protective devices which are capable of detecting and removing overcurrent conditions based on, e.g., tripping curves.

[0036] In yet other exemplary embodiments, a low voltage power distribution system may provide system control, including control of subsystem states, lighting items, circuit protective devices, and power source recovery. For example, the system may provide control of system and subsystem states. As another example, the system may provide operational control of electronic circuit breakers by, e.g., allowing, via an interface with flight deck avionics (FDA), for automatic or manual control of the state of one or more solid state power controllers (SSPCs) present in the SPDAs. As yet another example, the system may provide a capability to reset failed power sources by, e.g., sending a reset signal to a faulty power source when commanded by a user of the aircraft activating a reset switch via a flight deck control panel.

[0037] In some exemplary embodiments, a low voltage power distribution system may determine a system status by, e.g., detecting and reporting its status to the FDA. For example, the system may gather a general system status from one or more components and report the general status to the FDA.

[0038] In other exemplary embodiments, a low voltage power distribution system may detect and report system faults. For example, the system may perform built-in testing (BIT) to monitor for and detect failures and report detected failures to the flight deck (e.g., FDA) for corrective action, maintenance, and/or pilot awareness. As another example, the system may monitor for and detect low voltage power system faults and report detected faults to the flight deck (e.g., FDA) for corrective action, maintenance, and/or pilot awareness.

[0039] In some exemplary embodiments, a low voltage power distribution system may support system maintenance by, e.g., enabling software maintenance and/or providing an interface for maintenance equipment to perform system maintenance tasks. For example, the system may provide built-in testing equipment (BITE) to facilitate maintenance identification of failing components. As another example, the system may provide on wing capability to set rates of electronic breakers by, e.g., configuring the rate of one or more SSPCs using a software package. As yet another example, the system may support software configuration reporting via, e.g., a communication protocol to the flight deck. As another example, the system may support on wing software upload by, e.g., providing field-loadable SPDA software.

[0040] In some embodiments, each of the separate primary power conversion devices are in turn connected, via separate switching networks, to electrically separate low-voltage buses (i.e., wherein the low-voltage buses are not interconnected, or wherein the low-voltage buses are electrically separable). Thus, there may be at least two high-voltage power sources, at least two primary power conversion devices, at least two switching networks, and at least two 2 low-voltage buses. The two or more separate (or separable) buses may each be connected, via separate channels, to a separate high-voltage power source via the at least two primary power conversion devices. Furthermore, the capability to configure the absence of interconnections between the channels avoids any propagation of issues from one bus to another after any given failure on any one given channel.

[0041] In some embodiments, one of the high-voltage power sources, or alternatively a different high-voltage power source, is connected to an alternate power conversion device, which acts as a backup power source for any (or all) of the buses. In some embodiments, each primary channel (e.g., a primary power source and primary step-down power converter) may have a corresponding separate alternate channel (e.g., an alternate power source and/or an alternate step-down converter for each primary channel). In other embodiments, at least one alternate channel may provide backup power to all primary channels. In some embodiments, at least one bus may be connected to a low voltage battery which is configured to provide backup power via at least one switching device, if needed. In other embodiments, two or more buses may be connected to a third bus via switching devices, and the third bus may be connected to a low voltage battery via a switching device, wherein the low voltage battery is configured to provide backup power, if needed. In some embodiments, the third bus may be connected to one of the two or more buses. In some embodiments, the third bus may further be connected to either a low-voltage battery configured to provide backup power via a switching device, an alternate power supply (e.g., an alternate power converter and optionally an alternate power source) as described herein, or both. In some embodiments, the alternate conversion device may be connected to each of the two or more buses via one or more additional separate switching networks. In some embodiments, the switching networks connected to each respective bus (from their primary power conversion device and/or from the alternate power conversion device) may be controlled by a controller based on logic that involves determining one or more failing primary power conversion devices, one or more primary power sources, or any other component of one or more primary power supplies. In some embodiments, the primary power conversion devices and their respective switching network(s) may be controlled by a first, or primary, controller based on, e.g., hardwire logic including wires and/or relays, and the alternate power conversion device and its switching network(s) may be controlled by a second, or alternate, controller based on, e.g., conditional logic software or electronics. In some embodiments, each primary power conversion device and respective switching network may be controlled by separate primary controllers based on, e.g., simple, hardwire logic. In some embodiments, the alternate power conversion device and its switching network(s) may be separately controlled by one or more individual controllers based on, e.g., hardwire logic or conditional logic. Hardwire logic may suffice for the primary controller(s), because a primary controller must simply be programmed to cause a corresponding switching network to open upon a detected failure thereby disconnecting the failing component from the rest of the system. For example, a primary controller may include relays and contactors, splices, a programmable logic device, or a field programmable gate array (FPGA). Alternate controller(s), however, may require more complex logic, such as conditional logic (e.g., implemented via conditional logic software or electronics) due to the various potential failures that can occur along any one of the primary power channels and the various connections/switching networks which must be opened and/or closed to provide alternate power and disconnect one or more failing components as a result of detection of at least one failure.

[0042] In exemplary embodiments, two or more primary DC-DC power converters may be responsible for providing power to two or more 28VDC buses during normal operation. Each of the two or more primary power converters may receive high voltage power from a one or a specific pair of high voltage batteries and convert it to a regulated lower voltage (e.g., 28 VDC) level to supply the loads connected to one of the two or more electrical busbars. In some embodiments, each primary converter may have a nominal power capacity of, e.g., 2.0 kW (71 A@28 VDC) and may supply up to 150% and 200% of its nominal rating for 2 minutes and 5 seconds, respectively.

[0043] In exemplary embodiments, at least an alternate DC-DC power converter may be responsible for the initial power up of the LVPS and/or for providing backup power to any two buses in case of failure of any one of the primary DC-DC power converters. In some embodiments, this alternate power converter may also be powered by one or a pair of high voltage batteries and

may provide regulated 28VDC power output. In some embodiments, the alternate power converter may have a nominal power capacity of, e.g., 2.8 kW (100 A@28 VDC) and may supply up to 150% and 200% of its nominal rating for 2 minutes and 5 seconds, respectively.

[0044] In exemplary embodiments, two or more primary power distribution boxes (PDBs) may receive low voltage power, each from a primary DC-DC power converter, and may distribute it to the aircraft loads through two or more independent and physically segregated busbars (e.g., one in each box). In some embodiments, these boxes also may house the system's power distribution contactors, relays, and thermal circuit breakers (TCBs). In some embodiments, each box may have provisions to accommodate, e.g., 25 TCBs.

[0045] In exemplary embodiments, along with the PDBs, two or more Secondary Power Distribution Assemblies (SPDAs) may also be used to distribute power to low voltage loads. In some embodiments, feeders coming from the PDBs may provide power to the SPDAs which may further distribute it to aircraft loads through solid state power controllers (SSPCs). In some embodiments, monitoring and control of the SSPCs may be performed through the interface with the FDA system. In some embodiments, each SPDA may contain 24 SSPCs with current ratings varying from, e.g., 5 to 15 amperes.

[0046] In exemplary embodiments, each of the two or more independent primary power distribution channels may be connected to three segregated pairs of high voltage batteries (e.g., CH1//CH4, CH2//CH5, CH3//CH6). In some embodiments, the alternate power converter (ALT DCDC) may be connected to the same high voltage pair of batteries as one of the primary power converters. In some embodiments, the primary power distribution channels may operate isolated from each other during normal operation and under single failure conditions.

[0047] In some embodiments, hardware components such as contactors, relays, and busbars may be installed inside the power distribution box of each power distribution channel. In some embodiments, the primary power contactors (PPCL, PPCR and PPCC) may be commanded by hardwired logic, while the alternate power contactors (APCL, APCR and APCC) may be commanded by the alternate controller (ALT CONTR).

[0048] In some embodiments, the alternate controller may interface with the FDA system through ARINC 429 and with the primary and alternate converters through CAN BUS. In some embodiments, ARINC 429 may also be used for communication between FDA and the SPDAs, through which SSPCs can be monitored and commanded from the flight deck. In some embodiments, besides CAN BUS, the alternate power converter may also send discrete fault signals to the alternate controller. In some embodiments, low power command signals may flow from the alternate controller to the PDBs for control of alternate power contactors (APCs).

[0049] In some embodiments, discrete signals may also be the communication link between the primary power distribution boxes and their associated primary power converters and from the PDBs to the alternate controller and alternate power converter.

[0050] In some embodiments, the alternate controller may read the bus voltage from each PDB and the reset relays may receive redundant power from different busbars to make their activation possible when an associated busbar is powered off.

[0051] In some embodiments, the aircraft mode and DC-DC reset switches may provide discrete signals to the power distribution boxes, while the Flight Control System may send discrete signals to the PDBs and the alternate controller.

[0052] In some embodiments, high voltage lines (positive and negative) may run from the high voltage batteries to the power converters. In some embodiments, low voltage lines (positive and negative) may connect the power converters and the PDBs. In some embodiments, SPDAs may be connected to corresponding buses through dedicated feeders.

[0053] In exemplary embodiments, the LVPS may interface with a high voltage system which may be responsible for providing power to the LVPS. In some embodiments, this power may be supplied by one, two, three, or more high voltage power channels. For example, a first high voltage

power system (HVPS) channel may be responsible for providing power to a first primary power converter. The interface between the first HVPS channel and the LVPS may include a pair of cables (positive and negative) connecting the HVPS channel to the first primary power converter. As a further example, second and third HVPS channels may be similarly connected to, e.g., a second primary converter and an alternate power converter, respectively. In some embodiments, two or more HVPS channels may be interconnected at the HVPS side.

[0054] In exemplary embodiments, a Flight Deck Control Panel may provide command signals so the LVPS can automatically configure itself to the desired operational state and/or recover from a failure in the power conversion system. For example, an aircraft mode switch may provide indication of the required operational state (e.g., “OFF”/“STBY”/“GND”/“FLY”) to the power distribution boxes through a discrete open/ground signal. As another example, a DC-DC converters reset switch may inform the DC-DC converters reset command to the power distribution boxes through a discrete open/ground signal.

[0055] In exemplary embodiments, a Flight Control System (FCS) may be responsible for determining when an aircraft is on the ground and stationary, and for sending this information to the LVPS. For example, a discrete open/ground signal may be sent by a FCS to indicate when the aircraft is on the ground and not moving. In some embodiments, this signal may be used by the LVPS to prevent all power converters from shutting down in the air in case of a failure or erroneous command of the aircraft mode switch.

[0056] In exemplary embodiments, all loads of the LVPS connected to any one of the busbars may be powered at regulated voltage levels by the power converters and may further have their circuits protected by circuit protective devices. For example, LVPS loads powered by TCBs may be directly connected to the power distribution boxes and may have their circuits protected according to the thermal circuit breaker tripping curves. As another example, LVPS loads powered by SSPCs may be connected to the SPDA modules and may have their circuits protected by solid state power controllers providing equivalent protection functions, through software, of the TCBs.

[0057] In exemplary embodiments, any interfaces between the LVPS and FDA may be performed through communication protocols. In some embodiments, data transfer necessary to trigger and display messages, status, and warnings, as well as monitoring and control of SSPCs, may be performed through such interfaces. As an example, communication may be established through ARINC 429. In some embodiments, voltage, current, load measurements, contactors' status, and detected fault/fail signals may be passed from the alternate controller to FDA. As another example, SPDAs may be connected to FDA through an ARINC 429 network. In some embodiments, information about the status of the SSPCs may be passed from the SPDAs to FDA, while control signals may be processed in FDA and sent to the SPDAs to command the SSPCs.

[0058] In exemplary embodiments, modes of operation of the LVPS may be related to the aircraft modes, and either may be commanded by an aircraft mode switch. For example, an aircraft powered down configuration may represent the LVPS when the aircraft mode switch is at an “OFF” position. In some embodiments, during this condition, no power may be available from the high voltage batteries to the DC-DC converters, all power contactors may be open, and the system may be considered completely unpowered. As another example, an aircraft standby configuration may represent a point when the high voltage batteries may start providing power to the LVPS converters. In some embodiments, no low voltage power may be available if none of the converters have received an enable signal. In some embodiments, during this configuration, although the converters may already be receiving high voltage power, the system contactors may still be kept open and all low voltage busbars may remain unpowered. As yet another example, low voltage power up may start when the aircraft mode switch is moved to a ground (“GND”) position. In some embodiments, at this moment, the alternate converter may receive an enable signal and may start providing regulated low voltage power through its output. In some embodiments, an alternate controller may sequentially command the alternate power contactors to close, so the low voltage

electrical buses may start powering up, as fed by the alternate converter. In some embodiments, the alternate converter may be the only converter powering all buses during this operation mode. In some embodiments, this may be performed to check the alternate converter's operation and availability during each power up sequence.

[0059] As a further example, when the aircraft mode switch is moved to a fly ("FLY") position, the primary converters may also receive an enable signal and may start their internal initialization sequence. In some embodiments, the primary power contactors may be commanded to close by one or more primary controllers. In some embodiments, during a few milliseconds, all converters may be paralleled through the alternate power contactors. In some embodiments, after the system has stabilized, the primary power contactors may be sequentially opened, thereby disconnecting the alternate converter from the busbars and allowing the three power distribution channels to operate independently and isolated from each other. In some embodiments, the alternate converter may be kept enabled and ready to power any two busbars in case of failures. In some embodiments, the LVPS may then assume its normal operation configuration. In some embodiments, this configuration may be kept throughout the whole flight until the aircraft is ready for a powering down sequence.

[0060] Upon a determination of a failing primary power conversion device, the combination of switching networks may be configured, via the primary and alternate controllers, such that certain switches comprising the combination of switching networks open and/or close to provide proper electrical connections such that alternate power from the alternate power conversion device is provided to the two or more buses while simultaneously cutting off the failing primary power conversion device. Some embodiments may include one or more low-voltage batteries that can be connected to one or more of the buses permanently or, alternatively, via additional switching networks.

[0061] In some embodiments, the alternate power conversion device acts as an alternate power source for any single bus of the two or more buses without causing any of the buses to become interconnected via the logic implemented by the controller(s) of the switching networks. In some embodiments, the alternate power conversion device may be configured to act as an alternate power source for two or more buses, particularly in the case where multiple primary power conversion devices fail. In such cases, the switching networks may cause the two or more buses to become interconnected, allowing the alternate power conversion device to simultaneously power at least two of the buses. In some embodiments, the alternate conversion device may be connected any of the two or more buses, thereby removing the prior art requirement of having separate alternate power supplies for each bus to ensure high availability. As a result, in some embodiments, the weight of the aircraft may also be reduced significantly. In some embodiments, the architecture of the switching networks is utilized to provide power to a corresponding bus via the alternate power conversion device. In some embodiments, the controller(s) used to control the switching networks may be configured to prevent connection of the alternate power conversion device to a bus when a short circuit failure is detected on that bus, thereby preventing a condition where a bus short circuit failure causes both the primary power conversion device and the alternate power conversion device to become inoperable due, e.g., to overcurrent. In some embodiments, this solution maintains a capability of the alternate power conversion device to be available to be used as an alternate power source for any other bus(es) following a short circuit on a given bus.

[0062] For example, should any one primary converter fail during flight, the alternate controller may promptly command the alternate power contactor of the failed power distribution channel to close. In some embodiments, this may bring back the power to the affected busbar after a brief power interruption caused by the failure. In some embodiments, the other one or more busbars may not experience any interruption as all two or more channels may still operate segregated from each other. In some embodiments, the alternate converter may have enough capacity to power all loads connected to the failed channel.

[0063] As another example, in case two primary converters fail during flight, the alternate controller, similar to the one converter fail scenario, may command the alternate power contactors of the failed distribution channels to close. In some embodiments, the alternate converter, therefore, may power the two busbars from the failed channels. In some embodiments, no load shed may be necessary as the alternate converter capacity may be enough to power all loads from these two busbars. In some embodiments, in this configuration, one power distribution channel may be kept segregated from the other one or more busbars, which may be powered by the same converter.

[0064] As yet another example, and in the unlikely event that three or more primary power converters fail during flight, the alternate controller combined with hardwired logic may only command to close the alternate power contactors of the first two failed channels. In some embodiments, this logic may be necessary as the alternate converter may not be capable of powering all three buses during flight. In some embodiments, the busbar of the last channel to fail may remain unpowered, but all critical loads required for continued safe flight and landing may be kept powered by the remaining two busbars supplied by the alternate converter. In some embodiments, additionally, as one of the busbars may be kept de-energized as a result of this failure condition, a reset of the failed converters may be allowed and made possible through a reset (“DC/DC RST”) switch. In some embodiments, should the reset attempt be successful, the recovered primary converter(s) may reconnect to the system and may re-establish power to its associated busbar.

[0065] As a further example, a busbar short-circuit may be handled differently from a converter failure by the LVPS. In some embodiments, as the short-circuit may usually be permanent and may damage other power sources connected to it, the alternate controller, associated with a failure indication from the primary converter of the affected channel, may determine whether the failure was caused by a short-circuit or not. In some embodiments, if the short-circuit is confirmed, the alternate controller may prevent the associated alternate power contactor from closing, therefore avoiding any damage to the alternate converter. In some embodiments, the bus short-circuit event may also lead to loss of one busbar, but all critical loads required for continued safe flight and landing may be kept powered by the remaining two power distribution channels unaffected by the event.

[0066] As a result, the power distribution system may provide at least single channel systems, as well as double channel systems, triple channel systems, and so on. As a further result, the power distribution system removes the need for aircraft diversion under at least single failure conditions, minimizes power feeders voltage drop, minimizes a likelihood of power interruptions simultaneously affecting multiple buses, minimizes a need to utilize double power feeders to supply user systems, provides at least three electrically and physically independent channels under both normal operation and single failure conditions, minimizes cascading failures due to channel independence, and minimizes the loads required.

[0067] Now turning to the Figures, FIG. 1 illustrates an exemplary power distribution system **100**, consistent with disclosed embodiments. As shown in FIG. 1, in some embodiments, the disclosed power distribution system **100** may comprise a first high voltage channel **115** connected to a first primary conversion device (e.g., primary converter) **105** which is in turn connected via a first switching network **120** to a first bus **110**. As further shown in FIG. 1, the power distribution system **100** may further comprise a second high voltage channel **130** connected to a second primary conversion device (e.g., primary converter) **135** which is in turn connected via a second switching network **150** to a second bus **140**. As further shown in FIG. 1, the power distribution system **100** may further comprise a third high voltage channel **195** connected to an alternate/standby power conversion device (e.g., alternate standby converter) **190** (conversion device and converter are used interchangeably throughout this disclosure) which is in turn connected via a third switching network **196** to both the first bus **110** and the second bus **140**. As further shown in FIG. 1, a first power distribution block **125** may comprise at least the first switching network **120** and the first

bus **110**, and a second power distribution block **145** may comprise at least the second switching network **150** and the second bus **140**. As further shown in FIG. **1**, a first controller **121** may control the first switching network **120**, a second controller **131** may control the second switching network **150**, and an alternate controller **151** may control the third switching network **196**. The third switching network **196** may include a first portion corresponding to the first power distribution block **125** and a second portion corresponding to the second power distribution block **145**. The first and second portions of the third switching network may each be controlled by the alternate controller **151**.

[0068] FIG. **2** shows another exemplary power distribution system **200**, consistent with disclosed embodiments. As shown in FIG. **2**, in some embodiments, the disclosed power distribution system **200** may comprise a first high voltage channel **215** connected to a first primary conversion device (e.g., primary converter) **205** which is in turn connected via a first switching network **220** to a first bus **210**. As further shown in FIG. **2**, the power distribution system **200** may further comprise a second high voltage channel **230** connected to a second primary conversion device (e.g., primary converter) **235** which is in turn connected via a second switching network **250** to a second bus **240**. As further shown in FIG. **2**, the power distribution system **200** may further comprise a third high voltage channel **260** connected to a third primary conversion device (e.g., primary converter) **265** which is in turn connected via a third switching network **280** to a third bus **270**. As further shown in FIG. **2**, the power distribution system **200** may further comprise a fourth high voltage channel **295** connected to an alternate/standby conversion device (e.g., alternate standby converter) **290** which is in turn connected via a fourth switching network **296** to each of buses **210**, **240**, and **270**. As further shown in FIG. **2**, a first power distribution block **225** may comprise at least the first switching network **220** and the first bus **210**, a second power distribution block **245** may comprise at least the second switching network **250** and the second bus **240**, and a third power distribution block **275** may comprise at least the third switching network **280** and the third bus **270**. As further shown in FIG. **2**, a first controller **221** may control the first switching network **220**, a second controller **231** may control the second switching network **250**, a third controller **241** may control the third switching network **280**, and an alternate controller **251** may control the fourth switching network **296**. The fourth switching network **296** may include a first portion corresponding to the first power distribution block **225**, a second portion corresponding to the second power distribution block **245**, and a third portion corresponding to the third power distribution block **275**. The first, second, and third portions of the fourth switching network **296** may each be controlled by the alternate controller **251**.

[0069] FIG. **3** shows an exemplary mode of an exemplary power distribution system **300**, consistent with disclosed embodiments. As shown in FIG. **3**, in some embodiments, the disclosed power distribution system **300** may comprise a first high voltage channel **315** connected to a first primary conversion device (e.g., primary converter) **305** which is in turn connected via a first switching network **320** to a first bus **310**. As further shown in FIG. **3**, the power distribution system **300** may further comprise a second high voltage channel **330** connected to a second primary conversion device (e.g., primary converter) **335** which is in turn connected via a second switching network **350** to a second bus **340**. As further shown in FIG. **3**, the power distribution system **300** may further comprise a third high voltage channel **360** connected to a third primary conversion device (e.g., primary converter) **365** which is in turn connected via a third switching network **380** to a third bus **370**. As further shown in FIG. **3**, the power distribution system **300** may further comprise a fourth high voltage channel **395** connected to an alternate/standby conversion device (e.g., alternate standby converter) **390** which is in turn connected via a fourth switching network **396** to each of buses **310**, **340**, and **370**. As further shown in FIG. **3**, a first power distribution block **325** may comprise at least the first switching network **320** and the first bus **310**, a second power distribution block **345** may comprise at least the second switching network **350** and the second bus **340**, and a third power distribution block **375** may comprise at least the third switching network

380 and the third bus **370**. As further shown in FIG. 3, the first, second, and third switching networks **320**, **350**, and **380** may be configured such that all three primary power conversion devices **305**, **335**, and **365** are separately and electrically connected to each of their respective buses **310**, **340**, and **370**, and the fourth switching network **396** may be configured such that the alternate power conversion device **390** is not electrically connected to any of buses **310**, **340**, or **370**. As further shown in FIG. 3, a first controller **321** may control the first switching network **320**, a second controller **331** may control the second switching network **350**, a third controller **341** may control the third switching network **380**, and an alternate controller **351** may control the fourth switching network **396**. The fourth switching network **396** may include a first portion corresponding to the first power distribution block **325**, a second portion corresponding to the second power distribution block **345**, and a third portion corresponding to the third power distribution block **375**. The first, second, and third portions of the fourth switching network **396** may each be controlled by the alternate controller **351**.

[0070] FIG. 4 shows another exemplary mode of an exemplary power distribution system **400**, consistent with disclosed embodiments. As shown in FIG. 4, in some embodiments, the disclosed power distribution system **400** may comprise a first high voltage channel **415** connected to a first primary conversion device (e.g., primary converter) **405** which is in turn connected via a first switching network **420** to a first bus **410**. As further shown in FIG. 4, the power distribution system **400** may further comprise a second high voltage channel **430** connected to a second primary conversion device (e.g., primary converter) **435** which is in turn connected via a second switching network **450** to a second bus **440**. As further shown in FIG. 4, the power distribution system **400** may further comprise a third high voltage channel **460** connected to a third primary conversion device (e.g., primary converter) **465** which is in turn connected via a third switching network **480** to a third bus **470**. As further shown in FIG. 4, the power distribution system **400** may further comprise a fourth high voltage channel **495** connected to an alternate/standby conversion device (e.g., alternate standby converter) **490** which is in turn connected via a fourth switching network **496** to each of buses **410**, **440**, and **470**. As further shown in FIG. 4, a first power distribution block **425** may comprise at least the first switching network **420** and the first bus **410**, a second power distribution block **445** may comprise at least the second switching network **450** and the second bus **440**, and a third power distribution block **475** may comprise at least the third switching network **480** and the third bus **470**. As further shown in FIG. 4, the second and third switching networks **450** and **480** may be configured such that the second and third primary power conversion devices **435** and **465** are separately and electrically connected to each of their respective buses **440** and **470**, however the first primary power conversion device is a failed primary conversion device **405**, and as a result, the first switching network **420** may be re-configured such that the first bus **410** is no longer electrically connected to the failed primary conversion device **405**. As further shown in FIG. 4, the fourth switching network **496** may be configured such that the alternate power conversion device **490** is electrically connected only to the first bus **410**. As a result, the first bus **410** may continue to receive power via the alternate conversion device **490** while simultaneously being electrically disconnected from the failing primary conversion device **405**. As further shown in FIG. 4, a first controller **421** may control the first switching network **420**, a second controller **431** may control the second switching network **450**, a third controller **441** may control the third switching network **480**, and an alternate controller **451** may control the fourth switching network **496**. The fourth switching network **496** may include a first portion corresponding to the first power distribution block **425**, a second portion corresponding to the second power distribution block **445**, and a third portion corresponding to the third power distribution block **475**. The first, second, and third portions of the fourth switching network **496** may each be controlled by the alternate controller **451**.

[0071] FIG. 5 shows yet another exemplary mode of an exemplary power distribution system **500**, consistent with disclosed embodiments. As shown in FIG. 5, in some embodiments, the disclosed

power distribution system **500** may comprise a first high voltage channel **515** connected to a first primary power conversion device (e.g., primary converter) **505** which is in turn connected via a first switching network **520** to a first bus **510**. As further shown in FIG. 5, the power distribution system **500** may further comprise a second high voltage channel **530** connected to a second primary conversion device (e.g., primary converter) **535** which is in turn connected via a second switching network **550** to a second bus **540**. As further shown in FIG. 5, the power distribution system **500** may further comprise a third high voltage channel **560** connected to a third primary conversion device (e.g., primary converter) **565** which is in turn connected via a third switching network **580** to a third bus **570**. As further shown in FIG. 5, the power distribution system **500** may further comprise a fourth high voltage channel **595** connected to an alternate/standby power conversion device (e.g., alternate standby converter) **590** which is in turn connected via a fourth switching network **596** to each of buses **510**, **540**, and **570**. As further shown in FIG. 5, a first power distribution block **525** may comprise at least the first switching network **520** and the first bus **510**, a second power distribution block **545** may comprise at least the second switching network **550** and the second bus **540**, and a third power distribution block **575** may comprise at least the third switching network **580** and the third bus **570**. As further shown in FIG. 5, the second switching network **550** and **580** may be configured such that the second primary power conversion device **535** is separately and electrically connected to its respective bus **540**, however the first primary power conversion device is a failed primary conversion device **505**, and as a result, the first switching network **520** may be re-configured such that the first bus **510** is no longer electrically connected to the failed primary conversion device **505**. Additionally, as further shown in FIG. 5, the third primary power conversion device is a failed primary conversion device **565**, and as a result, the third switching network **580** may be further re-configured such that the third bus **570** is no longer electrically connected to the failed primary conversion device **565**. As further shown in FIG. 5, the fourth switching network **596** may be configured such that the alternate power conversion device **590** is electrically connected to both the first bus **510** and the third bus **570**. As a result, the first bus **510** and the third bus **570** each may continue to receive power via the alternate power conversion device **590** while simultaneously each of first bus **510** and third bus **570** may be electrically disconnected from the failing primary conversion devices **505** and **565**. As further shown in FIG. 5, a first controller **521** may control the first switching network **520**, a second controller **531** may control the second switching network **550**, a third controller **541** may control the third switching network **580**, and an alternate controller **551** may control the fourth switching network **596**. The fourth switching network **596** may include a first portion corresponding to the first power distribution block **525**, a second portion corresponding to the second power distribution block **545**, and a third portion corresponding to the third power distribution block **575**. The first, second, and third portions of the fourth switching network **596** may each be controlled by the alternate controller **551**.

[0072] FIG. 6 shows yet another exemplary mode of an exemplary power distribution system **600**, consistent with disclosed embodiments. As shown in FIG. 6, in some embodiments, the disclosed power distribution system **600** may comprise a first high voltage channel **615** connected to a first primary power conversion device (e.g., primary converter) **605** which is in turn connected via a first switching network **620** to a first bus **610**. As further shown in FIG. 6, the power distribution system **600** may further comprise a second high voltage channel **630** connected to a second primary conversion device (e.g., primary converter) **635** which is in turn connected via a second switching network **650** to a second bus **640**. As further shown in FIG. 6, the power distribution system **600** may further comprise a third high voltage channel **660** connected to a third primary conversion device (e.g., primary converter) **665** which is in turn connected via a third switching network **680** to a third bus **670**. As further shown in FIG. 6, the power distribution system **600** may further comprise a fourth high voltage channel **695** connected to an alternate/standby power conversion device (e.g., alternate standby converter) **690** which is in turn connected via a fourth switching

network **696** to each of buses **610**, **640**, and **670**. As further shown in FIG. **6**, a first power distribution block **625** may comprise at least the first switching network **620** and the first bus **610**, a second power distribution block **645** may comprise at least the second switching network **650** and the second bus **640**, and a third power distribution block **675** may comprise at least the third switching network **680** and the third bus **670**. As further shown in FIG. **6**, the first, second, and third primary power conversion devices are failed primary conversion devices **605**, **635**, and **665**, and as a result, the first, second, and third switching networks **620**, **650**, and **680** may be re-configured such that the first, second, and third buses **610**, **640**, and **670** are no longer electrically connected to the failed primary conversion devices **605**, **635**, and **665**. Additionally, as further shown in FIG. **6**, the fourth switching network **696** may be configured such that the alternate power conversion device **690** is electrically connected to both the first bus **610** and the second bus **640**. As a result, the first bus **610** and the second bus **640** each may continue to receive power via the alternate power conversion device **690** while simultaneously each of first bus **610**, the second bus **640**, and the third bus **670** may be electrically disconnected from the failing primary conversion devices **605**, **635**, and **665**. As further shown in FIG. **6**, a first controller **621** may control the first switching network **620**, a second controller **631** may control the second switching network **650**, a third controller **641** may control the third switching network **680**, and an alternate controller **651** may control the fourth switching network **696**. The fourth switching network **696** may include a first portion corresponding to the first power distribution block **625**, a second portion corresponding to the second power distribution block **645**, and a third portion corresponding to the third power distribution block **675**. The first, second, and third portions of the fourth switching network **696** may each be controlled by the alternate controller **651**.

[0073] FIG. **7** shows yet another exemplary mode of an exemplary power distribution system **700**, consistent with disclosed embodiments. As shown in FIG. **7**, in some embodiments, the disclosed power distribution system **700** may comprise a first high voltage channel **715** connected to a first primary conversion device (e.g., primary converter) **705** which is in turn connected via a first switching network **720** to a first bus **710**. As further shown in FIG. **7**, the power distribution system **700** may further comprise a second high voltage channel **730** connected to a second primary conversion device (e.g., primary converter) **735** which is in turn connected via a second switching network **750** to a second bus **740**. As further shown in FIG. **7**, the power distribution system **700** may further comprise a third high voltage channel **760** connected to a third primary conversion device (e.g., primary converter) **765** which is in turn connected via a third switching network **780** to a third bus **770**. As further shown in FIG. **7**, the power distribution system **700** may further comprise a fourth high voltage channel **795** connected to an alternate/standby conversion device (e.g., alternate standby converter) **790** which is in turn connected via a fourth switching network **796** to each of buses **710**, **740**, and **770**. As further shown in FIG. **7**, a first power distribution block **725** may comprise at least the first switching network **720** and the first bus **710**, a second power distribution block **745** may comprise at least the second switching network **750** and the second bus **740**, and a third power distribution block **775** may comprise at least the third switching network **780** and the third bus **770**. As further shown in FIG. **7**, the second and third switching networks **750** and **780** may be configured such that the second and third primary power conversion devices **735** and **765** are separately and electrically connected to each of their respective buses **740** and **770**; however, the first switching network **720** may be configured such that the first primary conversion device **705** is disconnected from a failing bus **710**, and the fourth switching network **796** may be configured such that the alternate power conversion device **790** is not electrically connected to any of buses **710**, **740**, or **770**. As a result, a minimum of two buses **740** and **770** remain operational while the failing bus **710** may be electrically disconnected from the power distribution system **700** in order to avoid propagation of issues associated with the failing bus **710**. As further shown in FIG. **7**, a first controller **721** may control the first switching network **720**, a second controller **731** may control the second switching network **750**, a third controller **741** may control the third

switching network **780**, and an alternate controller **751** may control the fourth switching network **796**. The fourth switching network **796** may include a first portion corresponding to the first power distribution block **725**, a second portion corresponding to the second power distribution block **745**, and a third portion corresponding to the third power distribution block **775**. The first, second, and third portions of the fourth switching network **796** may each be controlled by the alternate controller **751**.

[0074] FIG. **8** shows yet another exemplary mode of an exemplary power distribution system **800**, consistent with disclosed embodiments. As shown in FIG. **8**, in some embodiments, the disclosed power distribution system **800** may comprise a first high voltage channel **815** connected to a first primary conversion device (e.g., primary converter) **805** which is in turn connected via a first switching network **820** to a first bus **810**. As further shown in FIG. **8**, the power distribution system **800** may further comprise a second high voltage channel **830** connected to a second primary conversion device (e.g., primary converter) **835** which is in turn connected via a second switching network **850** to a second bus **840**. As further shown in FIG. **8**, the power distribution system **800** may further comprise a third high voltage channel **860** connected to a third primary conversion device (e.g., primary converter) **865** which is in turn connected via a third switching network **880** to a third bus **870**. As further shown in FIG. **8**, the power distribution system **800** may further comprise a fourth high voltage channel **895** connected to an alternate/standby conversion device (e.g., alternate standby converter) **890** which is in turn connected via a fourth switching network **896** to each of buses **810**, **840**, and **870**. As further shown in FIG. **8**, a first power distribution block **825** may comprise at least the first switching network **820** and the first bus **810**, a second power distribution block **845** may comprise at least the second switching network **850** and the second bus **840**, and a third power distribution block **875** may comprise at least the third switching network **880** and the third bus **870**. As further shown in FIG. **8**, the first, second, and third switching networks **820**, **850**, and **880** may be configured such that the first, second, and third primary power conversion devices **805**, **835**, and **865** are electrically disconnected from each of their respective buses **810**, **840**, and **870**, and the fourth switching network **896** may be configured such that the alternate power conversion device **890** is electrically connected to all of buses **810**, **840**, and **870**. As a result, all three buses **810**, **840**, and **870** remain operational in, e.g., a standby or ground mode of the aircraft. As further shown in FIG. **8**, a first controller **821** may control the first switching network **820**, a second controller **831** may control the second switching network **850**, a third controller **841** may control the third switching network **880**, and an alternate controller **851** may control the fourth switching network **896**. The fourth switching network **896** may include a first portion corresponding to the first power distribution block **825**, a second portion corresponding to the second power distribution block **845**, and a third portion corresponding to the third power distribution block **875**. The first, second, and third portions of the fourth switching network **896** may each be controlled by the alternate controller **851**.

[0075] FIG. **9** shows an exemplary system **900** for powering a low voltage power system, consistent with disclosed embodiments. As shown in FIG. **9**, a high voltage (HV) system **910** may provide high voltage (HV) power to the low voltage power system (LVPS) **930**. As further shown in FIG. **9**, a control interface **905** may provide information on aircraft mode and DC/DC reset and switch positions to the LVPS **930** which are used to control power up and power down of the aircraft and recover from DC/DC converter faults, respectively. As further shown in FIG. **9**, the Flight Control System (FCS) **915** may provide an On Ground Stationary (OGS) signal to the LVPS **930** which is used by the controllers to avoid power down in flight. As further shown in FIG. **9**, the Flight Deck Avionics (FDA) **920** may provide Solid State Power Controller(s) (SSPCs) commands to the LVPS **930** which are used to turn individual loads on and off, and the LVPS **930** may provide SSPCs statuses to the FDA **920**. As further shown in FIG. **9**, the LVPS **930** may provide low voltage (LV) power to LV systems **925** of the aircraft.

[0076] FIG. **10** shows an exemplary method **1000** for operating an exemplary power distribution

system, consistent with disclosed embodiments. At step **1010**, the method **1000** may comprise providing at least a first power source (e.g., a high voltage power source) connected to a first power converter (e.g., a step-down low voltage converter), the first power converter being connected to a first bus (e.g., a low voltage bus) via a first switching device. At step **1020**, the method **1000** may comprise providing at least a second power source (e.g., a high voltage power source) connected to a second power converter (e.g., a step-down low voltage converter), the second power converter being connected to a second bus (e.g., a low voltage bus) via a second switching device. At step **1030**, the method **1000** may comprise providing at least an alternate power converter (e.g., a step-down low voltage converter), the alternate power converter being connected to the first and second buses via at least a third switching device. The first power converter, the first bus, and the first switching device may be electrically separate or electrically separable from the second power converter, second bus, and second switching device. Furthermore, the alternate power converter and at least third switching network may be electrically separate or electrically separable from both the first and second converters and switching devices. At step **1040**, the method **1000** may comprise controlling the first and second switching devices via at least a first controller. At step **1050**, the method **1000** may comprise controlling the at least third switching device via an alternate controller. As a result of the operation of the first and alternate controllers in controlling the first, second, and at least third switching devices, the power distribution system is configured to allow the alternate power converter to act as a backup power supply while also interrupting any connection to a failing component of the power distribution system.

[0077] FIG. **11** shows another exemplary method **1100** for operating, via primary and alternate controllers, an exemplary power distribution system, consistent with disclosed embodiments. At step **1105**, the method **1100** may comprise operating with normal primary power without any failing power conversion devices. At step **1110**, the method **1100** may comprise detecting a first failing primary power conversion device. At step **1120**, the method **1100** may comprise opening a first primary switching network as controlled by a first controller in order to allow for step **1140**, which may comprise disconnecting the first failing primary power conversion device from the first bus. At step **1125**, the method **1100** may comprise closing a corresponding first portion of the alternate switching network as controlled by an alternate controller in order to allow for step **1145**, which may comprise connecting the alternate power conversion device to the first bus. At step **1115**, the method **1100** may comprise detecting a second failing primary power conversion device. At step **1130**, the method **1100** may comprise opening a second primary switching network as controlled by a second controller in order to allow for step **1150**, which may comprise disconnecting the second failing primary power conversion device from the second bus. At step **1135**, the method **1100** may comprise closing a corresponding second portion of the alternate switching network as controlled by the alternate controller (or by a second alternate controller) in order to allow for step **1160**, which may comprise connecting the alternate power conversion device to the second bus.

[0078] The foregoing description has been presented for purposes of illustration. It is not exhaustive and does not limit the invention to the precise forms or embodiments disclosed. Modifications and adaptations of the invention will be apparent to those skilled in the art from consideration of the specification and practice of the disclosed embodiments of the inventions disclosed herein.

[0079] The features and advantages of the disclosure are apparent from the detailed specification, and thus, it is intended that the appended claims cover all systems and methods falling within the true spirit and scope of the disclosure. As used herein, the indefinite articles “a” and “an” mean “one or more.” Similarly, the use of a plural term does not necessarily denote a plurality unless it is unambiguous in the given context. Words such as “and” or “or” mean “and/or” unless specifically directed otherwise. Further, since numerous modifications and variations will readily occur from studying the present disclosure, it is not desired to limit the disclosure to the exact construction and

operation illustrated and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the disclosure.

[0080] Other embodiments will be apparent to those skilled in the art from consideration of the specification and practice of the implementations disclosed herein. It is intended that the architectures and circuit arrangements shown in figures are only for illustrative purposes and are not intended to be limited to the specific arrangements and circuit arrangements as described and shown in the figures. It is also intended that the specification and examples be considered as exemplary only, with the true scope and spirit of the invention being indicated by the following claims.

Claims

1.-20. (canceled)

21. A power distribution system for an aircraft, comprising: a first power source connected to a first converter, the first converter being separably connected to a first bus, the first bus being connected to an alternate switching device; a second power source connected to a second converter, the second converter being separably connected to a second bus, the second bus being connected to the alternate switching device; an alternate power source connected to an alternate converter, the alternate converter being connected to the alternate switching device; wherein: the separable connection between the first converter and the first bus and the separable connection between the second converter and the second bus are controllable by at least one first controller, the at least one first controller being configured to operate using wire or relay logic; and the alternate switching device is controllable by at least one second controller, the at least one second controller being configured to operate using conditional logic.

22. The power distribution system of claim 21, wherein the alternate switching device is configured to electrically separate the first bus and the second bus based on a detected failure of a first load connected to the first bus or the second bus.

23. The power distribution system of claim 21, wherein the at least one second controller is further configured to modify the alternate switching device based on a second detected failure of a second load connected to the first bus or the second bus.

24. The power distribution system of claim 23, wherein the first load or the second load is associated with an engine of the aircraft.

25. The power distribution system of claim 21, wherein the alternate switching device includes two or more switches.

26. The power distribution system of claim 25, wherein the first bus is connected to a first switch of the alternate switching device, and the second bus is connected to a second switch of the alternate switching device.

27. The power distribution system of claim 26, wherein the at least one second controller is configured to, based on the conditional logic, open the first switch and close the second switch, or vice versa, upon a detected failure of the power distribution system.

28. The power distribution system of claim 21, wherein each of the separable connection between the first converter and the first bus and the separable connection between the second converter and the second bus includes at least one switch.

29. The power distribution system of claim 21, wherein the system comprises the at least one first controller and the at least one second controller.

30. The power distribution system of claim 21, wherein the at least one first controller is configured to, based on the wire or relay logic, cause the separable connection between the first converter and the first bus or the separable connection between the second converter and the second bus to open upon a detected failure of a load connected to the first bus or a detected failure of a load connected to the second bus.

- 31.** The power distribution system of claim 21, wherein the at least one first controller comprises a relay and contactor, one or more splices, a programmable logic device, or a field programmable gate array (FPGA).
- 32.** The power distribution system of claim 21, wherein the at least one second controller is configured to operate using conditional logic software.
- 33.** The power distribution system of claim 21, wherein the at least one first controller or the at least one second controller is configured to operate based further on data received from a flight control system or data based on an aircraft mode.
- 34.** The power distribution system of claim 21, wherein the alternate converter is configured to, via the alternate switching device, provide alternate power from the alternate power source to a first load connected to the first bus and to a second load connected to the second bus.
- 35.** The power distribution system of claim 21, wherein the first converter is electrically isolated from the first bus and the second converter is electrically isolated from the second bus.
- 36.** The power distribution system of claim 21, wherein the alternate power source is an only alternate power source of the system.
- 37.** The power distribution system of claim 21, wherein the at least one first controller or the at least one second controller is configured to prevent connection of the alternate converter to the first bus or the second bus upon a detected short circuit associated with the first bus or the second bus.
- 38.** A method for power distribution in an aircraft, the method comprising: connecting a first power source to a first converter, the first converter being separably connected to a first bus, the first bus being connected to an alternate switching device; connecting a second power source to a second converter, the second converter being separably connected to a second bus, the second bus being connected to the alternate switching device; and connecting an alternate power source to an alternate converter, the alternate converter being connected to the alternate switching device; wherein: the separable connection between the first converter and the first bus and the separable connection between the second converter and the second bus are controllable by at least one first controller, the at least one first controller being configured to operate using wire or relay logic; and the alternate switching device is controllable by at least one second controller, the at least one second controller being configured to operate using conditional logic.
- 39.** The method of claim 38, wherein: the alternate converter is configured to, via the alternate switching device, provide alternate power from the alternate power source to a first load connected to the first bus and to a second load connected to the second bus; and the alternate power source is an only alternate power source.
- 40.** The method of claim 38, wherein the at least one first controller or the at least one second controller is configured to prevent an electrical connection of the alternate converter to the first bus or the second bus based on a detected short circuit associated with the first bus or the second bus.
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